

Algebra 1
Unit 7
Lesson 8 Practice

Name Answer key
Date _____
Period _____

For questions 1-2, consider the expression $36x^2 - 64y^2$.

1. Rewrite the expression as a difference of two squares.

$$(6x)^2 - (8y)^2$$

2. Factor the expression.

$$(6x + 8y)(6x - 8y)$$

For questions 3-4, consider the expression $121m^2 - 25n^2$.

3. Rewrite the expression as a difference of two squares.

$$(11m)^2 - (5n)^2$$

4. Factor the expression.

$$(11m - 5n)(11m + 5n)$$

For questions 5-8, determine whether the quadratic expression is a difference of two squares. Then, factor the expression.

5. $49s^2 - 144m^2$

Yes

$(7s + 12m)(7s - 12m)$

6. $49g^2 + 84g + 36$

No

$(7g + 6)^2$

7. $9 - 4r^2$

Yes

$(2r + 3)(-2r + 3)$

8. $-100 + w^2$

Yes

$(w + 10)(w - 10)$

For questions 9-10, factor the expression.

9. $x^2 - 4x + 4$

$(x - 2)^2$

10. $81v^2 - 9p^2$

$9(9v^2 - p^2)$

$9(3v - p)(3v + p)$